

PROFILE SUMMARY

Data Engineer / Software Engineer with 2.6 years of professional experience (5+ years coding). I build reliable data pipelines and cloud-native systems across AWS and open-source stacks. Hands-on with streaming (Kafka, Spark Structured Streaming), orchestration (Airflow, dbt), data warehousing (Snowflake/BigQuery/PostgreSQL), and scalable services on Kubernetes and serverless. Strong focus on ELT/ETL design, data modelling, and production observability.

SKILLS

- Data & Big Data:** Apache Spark (SQL/Structured Streaming), Apache Kafka, Delta/Parquet, dbt, Airflow
- Cloud & DevOps:** AWS (S3, Lambda, API Gateway, DynamoDB, Cognito, SNS, CloudWatch, IAM), Docker, Kubernetes, GitLab CI/CD
- Warehousing & Storage:** Snowflake, Google BigQuery, PostgreSQL/PostGIS, MySQL, SQL Server, MongoDB, DynamoDB
- Backend & APIs:** Python (FastAPI), Java (Spring Boot), REST, Microservices, GraphQL
- Programming:** Python, Java, Kotlin, C++, JavaScript/TypeScript, SQL, R
- Testing & Quality:** JUnit, Espresso/Mockito, SonarQube, PIT (Mutation Testing)
- Tools:** Git/GitHub, Postman, Gradle, Terraform (basics), AWS SDK (Boto3), Amplify

PROJECTS

- BirdTag (Cloud) — Serverless Media System for Tagging & Search**
AWS: S3, Lambda, API Gateway, DynamoDB, Cognito, SNS; Python; REST | GitHub | Website
 - Designed event-driven ingestion with S3 triggers; Lambda functions generate thumbnails, extract metadata, and persist tags in DynamoDB (schema keyed by `media_id/tag_type`).
 - Built REST APIs (API Gateway + Lambda) for search/browse; integrated Cognito for auth and SNS for upload notifications.
 - Optimized cold-starts and memory/timeout settings; added CloudWatch metrics and alerts to improve reliability.
- Big Data Pipelines — Streaming & Batch Analytics**
Spark Structured Streaming, Kafka, Python, Parquet/Delta, Airflow/dbt (concepts) | GitHub
 - Implemented Kafka producers (multi-topic weather/energy) and a Spark consumer performing 6h/daily aggregations (windowing, watermarks) to Parquet/Delta sinks.
 - Engineered schemas and partitioning strategies for efficient reads; added job checkpoints and idempotent writes.
 - Outlined Airflow DAGs + dbt models for ELT in warehouse; added monitoring hooks and backfill strategies.
- PTV Bus Accessibility Analysis — Spatial Data Engineering**
PostgreSQL/PostGIS, GTFS, QGIS, Python/SQL | GitHub
 - Ingested GTFS and ABS boundaries; created spatial indexes and LGA/SA1-level layers.
 - Computed 400m catchments for commercial/education/hospital services; ran spatial joins (e.g., `ST_Buffer`, `ST_Intersects`, `ST_Union`) for coverage metrics.
 - Produced LGA rankings and choropleths for bus accessibility; documented pipeline and SQL for reproducibility.
- MultiAgent Route & Weather System — Tool-Calling Orchestration**
Python, FastAPI (tools), Autogen-style agents, Google Maps APIs, PostGIS | GitHub
 - Built agents that call routing + accident-count tools (PostGIS) and sample weather along polylines; returns polylines, legs, and filtered risk metrics.
 - Added tool registration & execution separation, robust error handling, and clean interfaces for future tools (e.g., road snapping, places).

EXPERIENCE

- Infor**

04/2022 – 07/2024

Associate Software Engineer — EPM Team
 - Automated deployments for reporting apps on Infor CloudSuite using **GitLab CI/CD**, **Docker**, and YAML pipelines across Linux environments.
 - Developed Python utilities to generate/archive report snapshots in **AWS S3**, improving auditability and recovery.
 - Managed infra with **EC2 ASG**, Launch Templates, ECR, Secrets Manager; implemented observability via **CloudWatch** and **SNS**.
 - Integrated with Infor Document Management (IDM); enforced least-privilege IAM and role-based access.
 - Participated in Agile sprints; coordinated releases with platform/ops to reduce lead time and deployment risk.

EDUCATION

- Monash University, Melbourne**

06/2024 – Present

Master of Information Technology
- Jawaharlal Nehru Technological University**

08/2018 – 06/2022

B.Tech. in Computer Science and Engineering